

ELEC 409 Final Assignment

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Executive Summary

In 1999, a team, Golub et al., showed that it was possible to use gene expression data to classify human acute leukemias. In an extension of this method, a new team of researchers, Pomeroy et al., performed a study from which they concluded that a patient's gene expression data could even be used to predict their clinical outcome. Their paper boasted that they had developed a model which could predict the clinical outcome of a medulloblastoma patient based on their gene expression data with nearly 80% accuracy. However, the data that this model was tested on was imbalanced with about 2/3 of the 60 patients being survivors. By examining the test data (which could only be found in the Supplementary Information extension of the paper), it is clear that the model was highly biased towards predicting patients as survivors. The model only predicted with 47.6% accuracy in cases where the patient did not survive. Since their model was biased towards the dominant class in the dataset, the high overall accuracy rating does not give an accurate representation of how predictable outcome really is from expression data.

In order to gain a better understanding of clinical outcome predictability from gene expression data, a re-analysis was done on the data used by Pomeroy et al. This time, a model was developed that optimized the discriminative capabilities for both classes rather than overall accuracy. By doing so, bias was reduced in prediction and better insight was given into the correlation of the data and the resulting clinical outcome. While the model was able to classify patients of both outcome correctly more often than not, the model had a low overall accuracy score. From these results, it can be deduced that predicting clinical outcome from gene expression data is much more complicated than the team behind Pomeroy et al. assert. The much less biased model shows very little correlation between expression data and clinical result. This is not to say that gene expression data cannot be used to predict clinical outcomes. However, there are many complex inter-genetic relationships that were not considered in Pomeroy et al.'s study and their assertion that the outcome is 'highly predictable' is an overstatement.

In order to derive a model with the best predicting capabilities possible, highly sophisticated techniques are needed that take into account the biological relationships between the genes in the data. Moreover, a much larger data set than 60 patients should be used when making such a claim. By optimizing a model for overall accuracy on such a small dataset, it is likely to find a method that overperforms the real best possible discriminator.

Introduction

Medulloblastoma is a cancerous tumor that starts as a small growth of cells in the lower-back region of the brain [1]. The tumor grows quickly, and the cancer can spread through a patient's cerebrospinal fluid. Although it is rare (an average of 357 people diagnosed yearly), Medulloblastoma is the most common form of brain cancer in children and has a 5-year survival rate of 72.1% [2].

In 2002, a team of researchers from several notable Universities published a paper, "Prediction of central nervous system embryonal tumor outcome based on gene expression" [3]. Borrowing from the 1999 paper "Molecular Classification of Cancer: Class Discovery and Class Prediction by Gene Expression Monitoring" [4], they used a method of predicting clinical outcome based on a patient's gene expression data. The researchers claimed that the clinical outcome of children with medulloblastoma was 'highly predictable'. The paper boasts out of the gene expression data and clinical results of 60 patients, their optimized model made only 13 classification errors, predicting the clinical outcome at about 80% accuracy. This paper failed to explain the bias their model showed towards predicting patients as survivors. It is only in the Supplementary Information extension of the paper that the team reveals that only 10 of the 21 deaths were correctly classified (47.6% accuracy). Although assessment using Fisher's Exact test did reveal correlation between the data and the model's classification, the model was less successful than described and reached a high margin of accuracy by favoring the dominant class.

Commented [WR1]: Fact check

The purpose of this assignment is to perform a re-analysis of the data used by Pomeroy et al. to study the predictability of medulloblastoma via gene expression data. This paper uses weighted voting of informative genes (as done by Golub et al.) to provide a better understanding of the clinical predictability with methods that could have been used at the time of the study. The model avoids the classification bias found in Pomeroy et al. by optimizing metrics more comprehensive than simply using prediction accuracy. Lastly, a machine learning algorithm was trained on the data optimizing the same metrics try to approximate the peak predictability that the gene expression data can provide.

Methodology

Prediction Goals

The main flaw in the study done by Pomeroy et al. is that the model favored accuracy above everything. In binary classification, if the data is disproportionately centered around one of the classes, solely optimizing prediction accuracy can lead to the majority class being favoured. This is the case in their study as the maximum total accuracy was reached when predicting survival in 80% of cases. Although this led to a favourable total accuracy, deaths were misclassified the majority of the time.

In order to reduce the bias in predictions, metrics that better favours discriminatory capabilities should be used for optimization. The area under the Receiver Operating Characteristics (ROC) curve is a better metric than accuracy for this because the ROC curve compares the true positive rate to the false positive rate. Taking the false positive rate into account is important for cases of imbalanced classes because it can be used to penalize over-favoring the dominant class. The area under the ROC curve provides a metric that quantifies the relationship between the true positive rate and false positive rate. By maximizing the area under the ROC curve the model's ability to discriminate between two classes is optimized. It should be noted that a higher area under the ROC curve still has the potential to

misrepresent data and does not ensure eliminating bias. A confusion matrix can be used auxiliary to area under the ROC scores to ensure that both positive and negative classifications are performing well.

Technology Used

The data analysis was done with Python using Jupyter Notebook. Python was chosen as it provides highly advance libraries specialized in data analysis and machine learning.

Library	Import	Purpose
numpy	numpy	Provides functions for handling mathematical computations with large arrays and matrices
panadas	pandas	Allows the data to be read in from the excel sheets in a manipulatable form
sklearn	StandardScalar	Standardizes data by removing the mean and scaling the data by dividing by the standard deviation
	RandomForestClassifier	Popular machine learning algorithm that avoids overfitting in smaller data sets; also provides functions for accessing feature importance once trained
	LeaveOneOut	Used to perform leave-one-out iterations by looping through data N times while removing a different data point each time
	roc_auc_score, precision_score, recall_score, f1_score, confusion_matrix, accuracy	Provides metrics for evaluating predicted and actual results

Procedure

Split Patient Data

The first step of the data analysis was splitting the data into a training set and a testing set. Before separating, 0s and 1s were added to the bottom of each patient column indicating if they patient survived. As specified in assignment instructions, both sets represented half of the total patients with no overlaps. The data was separated by randomly selecting half of the surviving patients and half of the deceased patients and moving them to an excel sheet label Training_Set. The remaining half of the data was moved to another excel sheet labeled Testing_Set. Moreover, the order of the data was randomized to disrupt any patterns in adjacent patients. Randomization was done by manual random selection.

Determine Gene Importance

Once the data was separated, the data was loaded into Jupyter Notebook via pandas. Then training data and class labels were parsed into a matrix and an array respectively. Using sklearn, the gene expression measurements in the training data were standardized. To determine the average feature importance, leave-one-out testing was used on the training data. For each set of N-1 gene-expression values and

corresponding outcomes, a sklearn random forest neural network was trained. For each of the leave-one out networks, the feature-importance was calculated using the model's built in 'feature_importances_' method. By doing so, N-1 feature importance values were collected for each gene-expression measurement. The average was taken of these feature importance values to calculate a final importance calculation for each gene. The higher the feature importance value for a given gene-expression measurement, the greater the contribution it had to the clinical outcome. Once the importance of each gene-expression measurement was calculated, it was possible to perform a weighted vote to predict a patient's clinical outcome.

Optimize Weighted Voting

Optimization technique

There were two parameters used to optimize the weighted voting model: the *number of genes* used, and the *threshold for determining the result* of the weighted voting. The optimal choices for these parameters were selected by rigorously evaluating different pairs of the two values on the original training set. The method was to perform a weighted vote with the 10 best informative genes, and analyse the result. Then, the number of top genes would be incremented by one and the vote would be taken again and this process repeated until it had been done for the top 75 genes. Moreover, for each set of genes tested, the weighted vote was performed using 20 evenly distributed threshold values. The result was 1300 weighted votes (*65 selections of # of genes x 20 thresholds each*). Every weighted vote was evaluated by calculating the area under its respective ROC curve. Out of the 1300 pairs, the AUC ROC score was optimized by including the top 49 genes and the best corresponding classification threshold.

Evaluation of Votes

To calculate the weighted votes, each of the informative genes used were multiplied by the corresponding feature importance value. By doing so, each vote was scaled proportionally to the importance of the gene. Once the weighted votes were calculated, 20 classification thresholds were chosen such that they were evenly distributed from the magnitude of the highest vote total to the lowest vote total between the 30 patients. In every iteration of this method, the AUC ROC score of the classifications was measured and the highest score was recorded.

Adjustments Made

To further analyze the success of the weighted voting model, the accuracy, F1 score, and confusion matrix were taken for the results of the optimized model. The original optimized model showed a low score of accuracy. Upon analysing the standardized expression values, it was found that the sum of the expression levels differed greatly from patient to patient. To counteract the effect this had on the vote, the data was normalized by dividing the gene-expression data by the sum of the used gene-expression data for each patient. After normalization, the optimization process was repeated resulting in a weighted vote that predicted clinical outcome with a much higher accuracy.

Despite normalization, there were still issues with the optimize weighted vote that needed to be fixed. The confusion matrix showed that even by optimizing AUC ROC score, the model was still highly biased. The highest AUC ROC score possible happened by classifying the vast majority of cases as survival. This problem was handled by adding further specifications to the optimization process. To ensure that both classes were being differentiated, any model that did not classify at least 40% of negative cases and 40% of positive cases correctly was eliminated. The optimization process was repeating targeting only

weighted votes that met this standard. Finally, an acceptable choice for the optimize model was found using the best 50 informative genes.

Results

Analysis

After creating a model with an optimized capability for discriminating both classes, a trial was done on the test data set. The model was fed the gene expression data for the remaining 30 patients and the results were as follows:

Accuracy: 60%

AUC ROC: 0.525

Precision: 0.706

Recall: 0.632

F1 Score: 0.667

Confusion Matrix:

	Predicted	
Actual	Failures	Survivors
Failures	6	5
Survivors	7	12

Beginning with the AUC ROC, the model achieved a score of 0.525. This was the highest AUC score out of 1300 tested models. A score of 0.525 shows that the model has very limited discriminative capabilities, just slightly better than a random guess. This notion is further backed by the 60% accuracy score the model achieved. The confusion matrix is where improvements over Pomeroy et al. are realized. While this model is less accurate when predicting patients of the survivor class (63%), it succeeds in predicting patients of the deceased class the majority of the time (54.5%, a 7% improvement).

While it is likely that a better computational method exists to predict these patients' clinical outcome, this weighted voting model is less biased than that of Pomeroy et al. By reducing bias in prediction results, the model's accuracy was greatly reduced. From this, it can be inferred that the predicting clinical outcome based on gene expression data is much more difficult than stated by Pomeroy et al. The relationship between genes is highly complicated and much more sophisticated analytics are needed to provide a prediction of worthy authority.

Fisher's Exact Test and Mathew's Coefficient Phi

Results of Fisher's Exact Test on the model:

one-tailed probability: 0.28703648

two-tailed probability: 0.45387687

These values are much greater than the 0.05 significance level. The results of this test informs that the null hypothesis should not be rejected. There is not enough evidence that the data is correlated.

Mathews Coefficient:

ϕ : 0.1358

The Mathews coefficient for the results shows that the model's predictions are better than random, but only by a small margin.

Shortcomings

While the model created in this study did improve upon the bias of the Pomeroy et al study, it is still far from perfect. To improve upon the findings of this report, there are more advance analytical methods that can be used to classify the data. Some ideas that can be used to achieve a better model for classifying the data include incorporating better methods to handle the class imbalance. Methods such as oversampling and using class weights can help deal with the disproportionate class sizes. As well, one could employ a more thorough optimization criteria rather than simply maximizing AUC ROC score. Using a mix of several different criteria could yield a more robust model. One last idea would be to use a more sophisticated method for identifying informative genes. While the method used does correctly calculate the average importance of each gene to the patient's clinical result, it does not account for any inter-gene relationships that may exist. Analyzing how certain gene expressions effect each other and performing feature reduction reducing the data set to orthonormal parameters would provide data untampered by counteracting gene relations.

Conclusion

The paper "Prediction of central nervous system embryonal tumor outcome based on gene expression" by Pomeroy et al makes the claim that the clinical outcome of a patient is highly predictable from their gene expression data. The paper fails to mention the bias in their model; it went unmentioned that their model predicted 80% of cases as survival and misclassified deceased patients more than half the time. In order to provide a better understanding of the data, a new analysis was done on the same gene expression data. This time a model was developed to maximize the accuracy of prediction for both classes, rather than the just overall accuracy score. The result was a model that classified the data correct more than half the time for both classes (survival and deceased), however it yielded a low score for accuracy overall. From these results, it can be concluded that prediction of clinical outcome from gene expression data is much more nuanced than it was made out to be by Pomeroy et al. With the greatest discriminatory capability that could be achieved, this model showed that the data had very little correlation towards clinical outcome. This is not to say that the gene expression data cannot be used to predict clinical outcome. However, it does show that the relationship between a patient's gene expression data and their clinical outcome is much too complex to predict with a basic weighted voting of k-means clustering models. Much more sophisticated methods are needed that take into account the biological relationships these genes have with each other and how their interaction can inform clinical outcome.

Bibliography

1. Mayo Foundation for Medical Education and Research. (2023, April 21). *Brain Tumor*. Mayo Clinic. <https://www.mayoclinic.org/diseases-conditions/medulloblastoma/cdc-20363524>
2. *Medulloblastoma diagnosis and treatment*. National Cancer Institute. (n.d.). <https://www.cancer.gov/rare-brain-spine-tumor/tumors/medulloblastoma>
3. Pomeroy et al. (2002, January 24). Prediction of central nervous system embryonal tumour outcome based on gene expression. *Nature*.
4. Golub et al. (1999). Molecular classification of cancer: Class Discovery and class prediction by Gene Expression Monitoring. *Science*, 286(5439), 531–537. <https://doi.org/10.1126/science.286.5439.531>